

Using the model version with a real landscape, vary the parameter q and see how it affects the interaction between landscape and butterfly movement. Start the butterflies in various places. From just looking at the View, what value of q seems best for finding mates on or near hilltops?

A q value of about 0.3 seems best for finding mates on or near hilltops. At high values of q , around 0.7 to 1.0, butterflies are more likely to move uphill and become trapped on nearby local hilltops. This creates many small, separated clusters rather than bringing butterflies together. At lower q values, around 0.4 and below, the butterflies move more randomly and are more likely to leave smaller local hilltops and eventually concentrate around larger, higher-elevation areas. A q of about 0.3 appeared to produce the largest clusters, which would make it easier for butterflies to encounter potential mates.